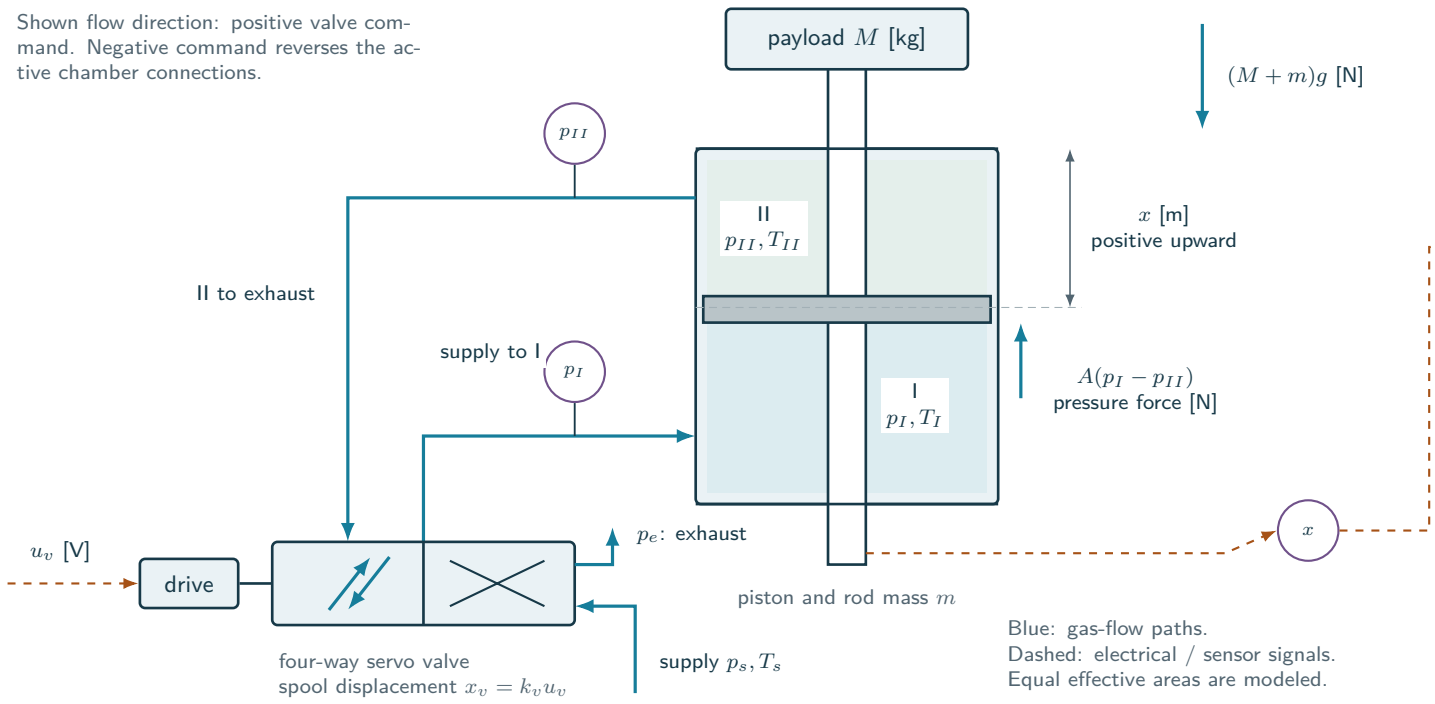
Vertical servo-pneumatic actuator

Paper section 3.5: *Servo-pneumatic actuator*

Two variable-volume chambers, valve flow and a vertical payload



Absolute physical variables

p_I, p_{II}: pressure [Pa].
T_I, T_{II}: temperature [K].
m_I, m_{II}: gas mass [kg].
A: effective piston area [m^2].

$V_I = V_0 + Ax$
$V_{II} = V_0 - Ax$
$p_j = m_j R T_j / V_j.$

Physical variant state

$[x, v, m_I, m_{II}, T_I, T_{II}]^\mathsf{T}$

Mass and energy balances evolve each chamber. Flow can choke or reverse. Valve and thermal parameters require calibration.

Vertical force balance

$(M+m)\dot{v} = A(p_I - p_{II}) - f(v) - (M+m)g - F_{ext}.$

The physical variant includes gravity explicitly.

Three model meanings

Eqs. (33)–(34): literal survey form.
Local identified models: deviation states with operating offsets removed.
Six-state model: separate physical variant.

Units and payload

Mass-specific $R = R_u / M_{air}$ [J/(kg K)].
Each physical simulation has constant payload; changing payload in motion needs additional assumptions.

Functional explanatory diagram, not a standardized pneumatic schematic. The survey's gravity omission and gas-constant units are discussed in PNEUMATIC_AUDIT.md. The physical valve closure is an explicitly parameterized teaching variant, not an identified replica of the apparatus.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.